

SAAIM ABDULLAH

Software Engineer

+92 324 9923289 saaim.abdullah.work@gmail.com

[linkedin.com/in/saaim-abdullah](https://www.linkedin.com/in/saaim-abdullah) github.com/saaim12 saaim.site medium.com/@saymmalik08

Summary

Software engineer with experience across backend development, frontend work, system architecture, and data engineering. I build and maintain scalable APIs, cloud infrastructure, and data pipelines, and I'm comfortable working end to end on a product, from designing the data model to shipping the interface and keeping it running in production.

Experience

Fitter Health

May 2026 – Jul 2026

Software Engineer

Spain/Remote (Contract)

- Built the backend for a preventive health-tech platform in **Django/DRF** over a 106-table **PostgreSQL** schema, covering membership credits, coach session booking, an e-commerce store, and an admin CRM, with a **Next.js** front end.
- Deployed on **AWS ECS** with RDS, S3, SES, SNS, and SQS; ran background and scheduled work on **Celery** and **Redis**, and integrated **Odoo** invoicing with payment status synced back to the platform database.
- Fixed silent message loss in the user re-engagement notification pipeline by buffering **SNS** fan-out through **SQS**, and handed the client a documented API surface backed by 135 passing tests.
- Designed and shipped an ETL pipeline into a **star-schema PostgreSQL** warehouse serving three consumers: ML feature workflows for the data science team, product performance analytics on best-selling items, and aggregated reporting for external partners.
- Automated member lifecycle email on **Event Bridge-scheduled Lambda** and **SES**, covering renewal reminders that replaced manual call-and-chase and inactivity re-engagement, and eliminated silent message loss by buffering **SNS** fan-out through **SQS**.

Expertflow – Cisco UCCE Solution Partner

Jun 2024 – Feb 2026

Software Engineer

Lahore, Pakistan

- Built and maintained backend **REST APIs** serving **1,000–3,000 daily users**, deployed as **Docker** containers on **AWS EC2** behind an **Application Load Balancer** with rate limiting, IAM, and VPC controls.
- Designed and shipped event-driven batch and streaming **ETL** pipelines on AWS using **Lambda**, **MWAA (Airflow)**, **EMR**, **PySpark**, Glue, and S3, automating ingestion, standardization, and cataloging of raw inputs into analytics-ready **Parquet** datasets for BI reporting and ML feature workflows.
- Developed a real-time voice/video communication SDK on **WebRTC** and the REST APIs binding frontend clients, SDK modules, and third-party platforms together.
- Implemented a chatbot workflow that cut first-response time on inbound queries, and delivered **SQL**-based analytics for customer insight reporting.
- Hardened platform access with time-bound permission controls, and worked across **Angular** frontend, backend, and SDK teams to ship end-to-end features.
- Reduced admin panel response time **130 ms to 30 ms** through **Redis** caching and database index tuning.
- Designed and deployed the backend as a **distributed system** on AWS EKS, splitting services across pods with Horizontal Pod **Autoscaler** policies that scaled replicas on request throughput to keep the system available under traffic spikes and improved overall **scalability**.

Projects

Real-Time E-Commerce ETL Pipeline | *Kafka, PySpark, Airflow* | [Article](#)

- Built a **medallion-architecture** (bronze/silver/gold) streaming pipeline moving Kafka-published orders into a **Parquet** bronze layer on S3-compatible storage via **PySpark Structured Streaming**. Batch and streaming Pipelines
- Modeled a gold **star schema** (customer, product, and date dimensions around a sales fact table) and orchestrated batch stages with an **Airflow DAG** on Docker Compose.

Multilingual, Multi-Tenant RAG Engine | *Django, PostgreSQL, RAG* | [Article](#)

- Built a production-style **RAG** backend with tenant-isolated retrieval, local multilingual embeddings, and grounded generation for a localization Q&A service on **Django** and **PostgreSQL 18**.
- Kept each tenant's corpus and retrieval path isolated at the query layer, and documented the setup, database friction, and fixes in a companion engineering article.

Autoencoder Compression for Edge Devices | *Python, TensorFlow, TF Lite Micro, Flask, AWS*

- Developed an autoencoder to compress and reconstruct sensor data for lightweight edge-device communication, tuning latent-vector size for compact representation.
- Converted the encoder for microcontroller-side inference and exposed the decoder through a **Flask API** on AWS EC2, with S3 integration for encoded payload handling.

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL

Backend & APIs: Django DRF, Celery, Node.js, Express.js, REST API Design, Event Driven Systems, Microservices

Data Engineering: ETL Pipelines, Apache Kafka, PySpark, Apache Airflow, AWS Glue, AWS EMR, Data Warehousing, Star Schema, Medallion Architecture

Cloud / DevOps: AWS (Lambda, EC2, ECS, S3, RDS, SQS, SNS, SES, IAM, VPC, ALB, EMR, EKS, Redshift), Docker, Kubernetes, GitLab

Databases: PostgreSQL, MongoDB, MsSql, DynamoDB, Redis

Frontend: React, Next.js, Angular

Education

COMSATS University Islamabad, Lahore Campus

Bachelor of Science in Computer Engineering

Lahore, Pakistan

Relevant coursework: Data Structures and Algorithms, Database Management, Computer Networks, Operating Systems, Software Engineering, Computer Architecture, Machine Learning.

Certifications

[AWS Cloud Technical Essentials](#) | [Introduction to Data Engineering](#) | [Getting Started with Git and GitHub](#) | [Claude Code](#)
101